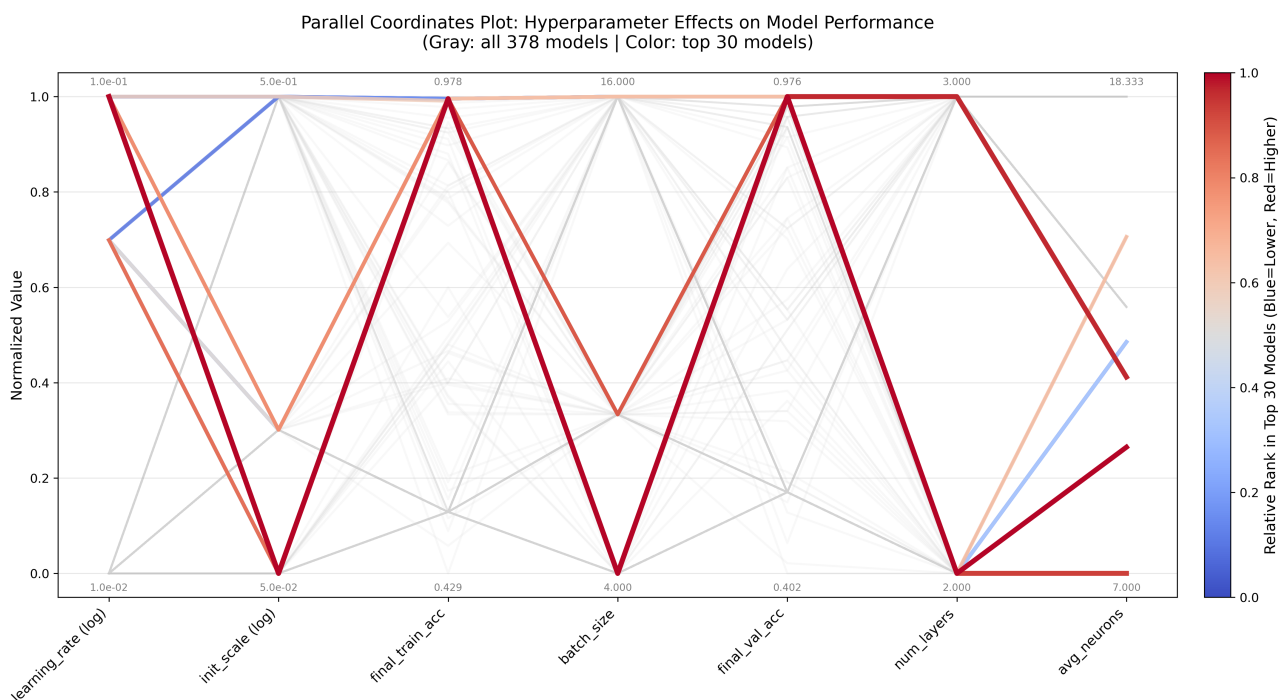


Hyperparameter optimization setup and sweep space. We conducted a grid-based hyperparameter sweep for an MLP classifier and evaluated performance on a held-out validation set. The search space was:

- **Hidden layer architectures (hidden_sizes):** ([7,7], [10,10], [15,15], [10,15], [10,15,10], [15,10,15], [30,15,10])
- **Learning rate (LR):** ({0.01, 0.05, 0.1})
- **Batch size (BS):** ({4, 8, 16})
- **Activation:** ({`\text{ReLU}`, `\text{sigmoid}`})
- **Initialization scale (init_scale):** ({0.05, 0.1, 0.5})

Results were summarized using a **Parallel Coordinates Plot (PCP)**: each polyline corresponds to one run, axes represent hyperparameters and final metrics, and color encodes validation performance (with a focus on highlighting the best-performing subset against all runs in gray).

Key PCP finding: performance is dominated by optimization-related hyperparameters (LR, init_scale, activation). The PCP shows that top-performing lines cluster tightly in specific regions of the hyperparameter space, while bottom-performing runs concentrate in a different region. This separation is most pronounced for **learning rate**, **initialization scale**, and **activation function**, suggesting these factors control trainability and convergence quality more strongly than raw model size.



Top-5 configurations reveal a consistent “high-performing recipe.” The **top 5** runs (all with **Val Acc = 0.97561**) share near-identical hyperparameters:

- **Activation: ReLU** in all **top-5** runs.
- **Initialization: init_scale = 0.05** in all **top-5** runs.

- **Learning rate:** predominantly **LR = 0.10** (4/5 runs), with one run at **LR = 0.05**.
- **Batch size:** mainly **BS = 4** (4/5 runs), with one run at **BS = 8**.
- **Architecture:** top performance is achieved by **multiple model sizes**, including [7,7], [10,10], and [10,15,10] (2–3 layers). This indicates that, once optimization is well-tuned (ReLU + small init + sufficiently large LR), **several architectures are capable of reaching the same high validation accuracy**.
- **Loss/fit quality:** the best model (ID 91) has **final train/val loss $\approx 0.0974/0.0978$** and **train/val acc = 0.9756/0.9756**, implying both low loss and excellent generalization.

Bottom-5 configurations highlight a failure mode driven by sigmoid + low LR (+ often large init). The **bottom 5** runs show a strikingly consistent pattern:

- **Activation:** **sigmoid** in all **bottom-5** runs.
- **Learning rate:** **LR = 0.01** in all **bottom-5** runs.
- **Initialization:** mostly **init_scale = 0.5** (4/5 runs), with one run at **init_scale = 0.1**.
- **Batch size:** biased toward **larger BS** (often **8 or 16**).
- **Architecture:** poor performance occurs across both moderate and large networks (e.g., [10,10], [15,15], [15,10,15], [30,15,10]), reinforcing that **capacity alone cannot compensate for unfavorable optimization settings**.
- **Loss/fit quality:** bottom runs have **final train/val loss $\approx 0.688\text{--}0.693$** , close to the typical cross-entropy value for near-random guessing in binary/multi-class settings, consistent with **training stagnation or underfitting**.

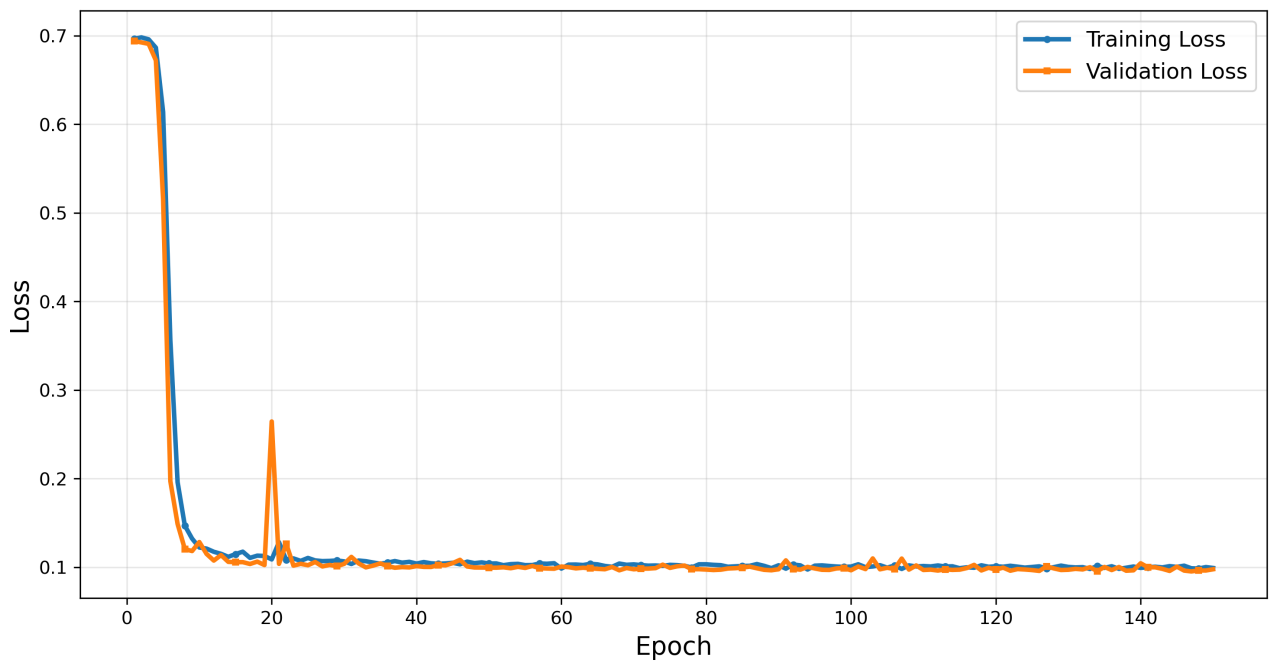
Interpretation of individual hyperparameters.

- **Learning rate (LR):** Higher LR (0.1, sometimes 0.05) is strongly associated with top performance, while LR=0.01 is associated with bottom performance. This suggests that **LR controls whether the model reaches a good minimum within the fixed epoch budget**.
- **Activation:** ReLU consistently appears in the top-5, whereas sigmoid appears in the bottom-5. This indicates that **ReLU provides more reliable gradient flow and optimization stability** in this task, while sigmoid frequently leads to **slow learning and saturation effects**.
- **Initialization scale (init_scale):** Small init_scale (0.05) is shared by all top-5 runs, while large init_scale (0.5) dominates the bottom-5. This implies that **large initial weights likely destabilize or saturate activations (especially with sigmoid), harming trainability**.
- **Batch size (BS):** Smaller batch sizes (especially BS=4) are overrepresented in the top set, whereas larger batch sizes (BS=8/16) are common among bottom runs. This suggests that **smaller batches may improve optimization dynamics and help reach better solutions**, particularly when coupled with a sufficiently large LR.
- **Architecture (hidden_sizes / depth):** Multiple architectures achieve the same best validation accuracy once optimization hyperparameters are favorable, indicating **architecture is not the primary bottleneck** here. Conversely, even large models can fail

under poor optimization settings, confirming that **trainability dominates capacity** in this sweep.

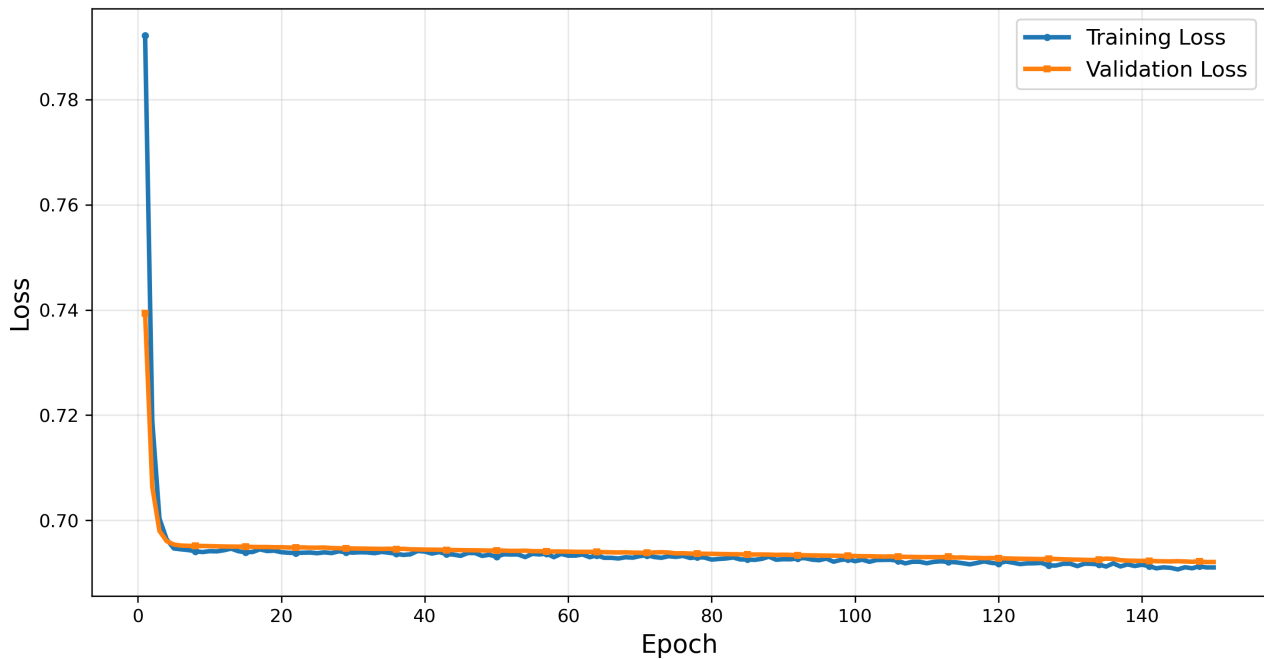
Best model (ID 91) loss curves and generalization. The best model uses **[10,10], ReLU, LR=0.1, BS=4**, and attains **Train Acc = 0.9756** and **Val Acc = 0.9756** (gap **0.0000**). Its loss curves show a **rapid early decrease** from ~ 0.69 to ~ 0.11 , followed by a stable low-loss plateau. Training and validation losses remain **nearly overlapping** throughout training (**0.0974 vs 0.0978**, loss gap **0.0004**), providing strong evidence of **excellent generalization with no observable overfitting** (a brief transient fluctuation in validation loss does not change the overall convergence pattern).

Best Model Loss Curves (Val Acc: 0.9756)
Architecture: [10, 10], LR: 0.1, BS: 4



Worst model (ID 180) loss curves and generalization. The worst model uses [10,15], sigmoid, LR=0.01, BS=16, init_scale=0.5 and achieves Train Acc = 0.4293 and Val Acc = 0.4024 (gap 0.0268). Its loss curves decrease slightly at the start but then plateau around ~0.69–0.70 with minimal improvement. The final training/validation losses are high and close (0.6903 vs 0.6921, loss gap 0.0018), indicating the model **does not fit the training data well** and generalizes poorly. This behavior is characteristic of **underfitting / optimization failure**, plausibly caused by the adverse combination of **sigmoid + large init_scale** (saturation/vanishing gradients) together with **low LR + large BS** (small effective parameter updates).

Worst Model Loss Curves (Val Acc: 0.4024)
Architecture: [10, 15], LR: 0.01, BS: 16



Overall conclusion. The PCP and the top/bottom configuration analysis collectively show that validation performance is primarily controlled by **optimization-related hyperparameters**: **ReLU + small init_scale (0.05) + sufficiently large LR (0.1, sometimes 0.05)** consistently yields near-optimal accuracy across multiple architectures, whereas **sigmoid + LR=0.01 (often with large init_scale=0.5)** leads to training stagnation and poor validation accuracy regardless of network size.